

The North-South Divide: Quantifying the Socioeconomic Drivers of England's Life Expectancy

1. Introduction

The increasing regional health divide between the North and South of England is currently an economic and humanitarian crisis. The Marmot Review 10 Years On confirmed that life expectancy improvement in England had stalled for the first time in a century, with the gap between the two regions continuing to grow substantially (Marmot et al., 2020). Recent statistics indicate a significant wealth and health gap, with men living in the poorest areas of Northern England likely to die 10.7 years younger than their counterparts living in the wealthiest areas of Southern England (Office for National Statistics, 2024). As well as social consequences, this disparity also leads to high economic costs. The Northern Health Science Alliance revealed that £13.2 billion annually is lost to the national economy due to the productivity losses associated with this health disparity, making it a top priority for policymakers (Bambra, Munford and Brown, 2018).

Existing research describes the North-South health inequality but often fails to identify the underlying socioeconomic drivers. Consequently, the policies aimed towards reducing this inequality remain inefficient and tend to fail due to a lack of evidence on which factors influence regional life expectancy (Rutter et al., 2017). This study aims to assess the severity of the life expectancy inequality between Northern and Southern England and identify which socioeconomic factors drive this disparity, while investigating the key predictors of regional inequality. The analysis seeks to provide evidence for governments and policymakers to develop focused initiatives where expenditure is used efficiently, targeting the specific causes of life expectancy inequality.

2. Analytical questions and data

2.1 Analytical Questions

To systematically achieve the research objectives and provide robust statistical evidence, this study will focus on four specific questions:

- What is the magnitude and significance of the North-South divide in life expectancy?
It is necessary to statistically confirm the size and significance of the differences in life expectancy as a baseline for when analysing the impact of the socioeconomic factors.
- Which IMD Sub-domains independently predict life expectancy?
To assess the extent to which each IMD sub-domain contributes independently to life expectancy, and to assess its relative significance.
- Does classifying regions more finely better explain life expectancy inequality?
To investigate spatial variation in life expectancy and evaluate whether specific micro-regions are responsible for the disparity.

2.2 Data Description

Cross-sectional data from 2019 was derived by merging five separate official datasets from the Office of National Statistics (ONS) and the UK government website. 2019 data were used to capture the regional inequalities in life expectancy, avoiding any temporary distortions caused by COVID-19.

Variable Name	Role	Type	Definition & Measurement
Area Code	ID	Nominal	Official ONS GSS code (e.g., E06000001)
Area Name	ID	Nominal	Name of the Local Authority (e.g., Hartlepool).
Latitude	Coordinate	Numerical	Geographic centroid (North-South position).
Longitude	Coordinate	Numerical	Geographic centroid (East-West position).
Region	Feature	Categorical	Official regions of the England

Mean_LifeExp	Target	Numerical	The mean life expectancy for each local authority in England (years)
IMD_Score	Predictor	Numerical	Overall weighted sum of the 7 subdomains
Income_Score	Predictor	Numerical	Proportion of the population that is income deprived (scale: 0-1) for each local authority in England
Employment_Score	Predictor	Numerical	Proportion of people in working age excluded from employment (scale: 0-1)
Education_Score	Predictor	Numerical	Lack of skills or education attainment (higher value = worse education deprivation)
Health_Score	Predictor	Numerical	Health deprivation or disability, factor score (scale >0 = higher health deprivation)
Crime_Score	Predictor	Numerical	Increased risk of crime, factor score (scale >0 = higher crime)
LivingEnvironment_Score	Predictor	Numerical	Housing and Air quality (higher value = worse environment)
Median_Age	Control	Numerical	Median age of the population of each local authority (Years)
Population_Density	Control	Numerical	Number of people per square km
Age 65+%	Control	Numerical	population aged 65 and over (%)

Table 1: Variables and their definitions and measurements

2.3 Potential Limitations

There are three limitations that shape the scope of this analysis. Using Local Authority Districts as the unit of study has the potential to mask variation in deprivation within the local areas, creating an overestimation of severe deprivation. The socioeconomic predictors are also measured at different levels; for example, income is expressed as a proportion, while crime is expressed as a factor score, which makes it difficult to interpret multivariable models and perform cluster analysis. Finally, the study will be cross-sectional and will establish a pre-COVID baseline is limited in its ability to make causal inferences between the independent variables and life expectancy.

3. Analysis

3.1 Data Wrangling

The Local Authority (LAD) code was used as the key for merging the datasets. The integrated dataset was then filtered out from any records with missing values and produced a final sample size of 304. A test for duplicates revealed a single distinct entry for each local authority, which created a clean and consistent dataset.

3.2 Exploratory Data Analysis (EDA)

Variable Name	Mean	Median	StDev
Mean_LifeExp	81.65	81.84	1.445
IMD_Score	19.83	18.63	7.995
Income_Score	0.117	0.109	0.045
Employment_Score	0.093	0.089	0.035
Education_Score	20.88	19.16	8.710
Health_Score	-0.115	-0.127	0.630
Crime_Score	-0.131	-0.134	0.514
LivingEnvironment_Score	20.57	19.08	8.695
Median_Age	42.21	42.35	5.142
Population_Density	1842	771.5	2693
Age 65+%	19.73	19.66	4.880

Table 2: Descriptive Statistics

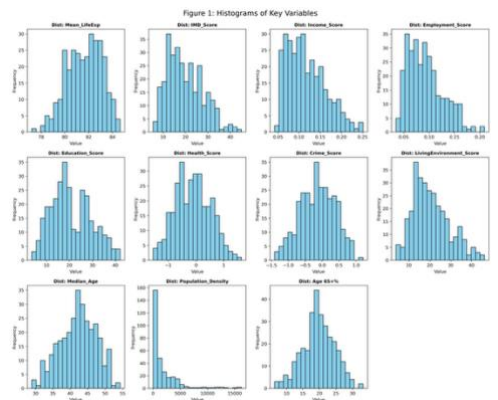


Figure 1: Histograms

In Table 2, the life expectancy has a standard deviation of 1.45 years; this indicates variation in life expectancy across LADs. The mean of Population_Density (1,842 people/km²) is twice its median of 771.5 people/km², indicating a strong right skew, suggesting high-density observations, visually confirmed in Figure 1. Figure 2 justifies the necessity to log-transform Population_Density to reduce skewness.

Figure 2: Boxplots

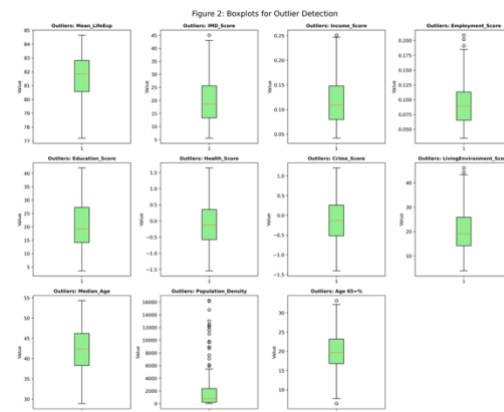
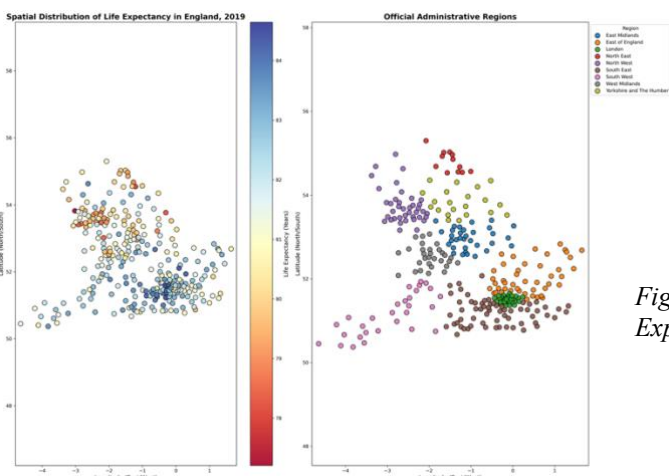


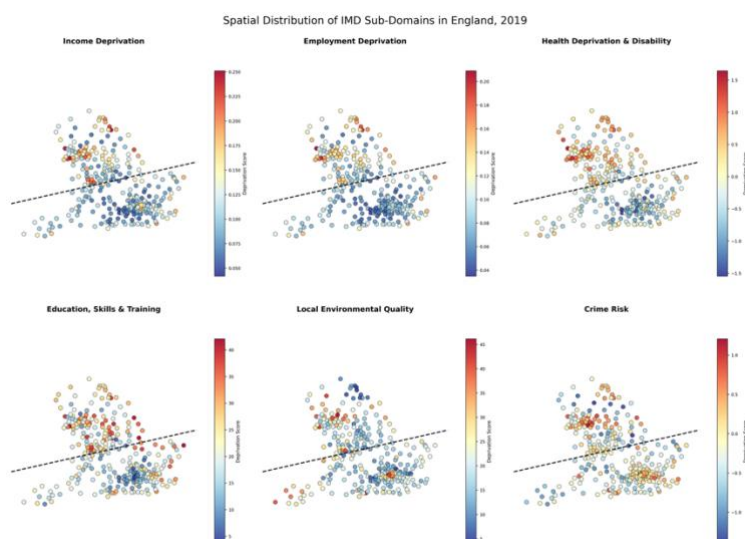
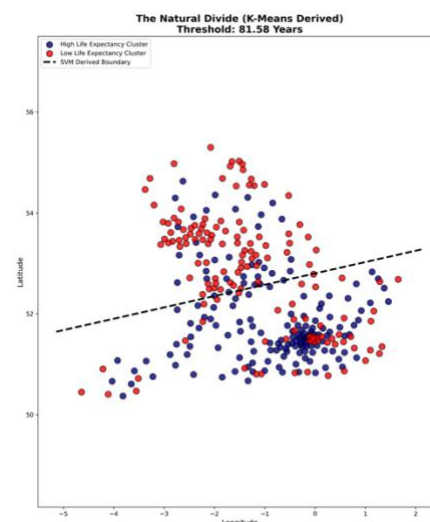
Figure 3 illustrates a significant latitudinal health gradient. The lowest Life Expectancy values (red) are located primarily within the Northwest region and the Northeast region. Conversely, there is an extensive area of higher Life Expectancy (blue) in both Southeastern and Home County regions, indicating that the North-South divide remains a major geographical characteristic of this dataset.

Figure 3: Scatter graph – Spatial Distribution of Mean Life Expectancy



Instead of using the mean or median life expectancy, a K-means clustering method was used to provide a data-based threshold (81.58 years). Using this threshold, the dataset was partitioned into LADs with low and high mean life expectancy. A linear Support Vector Machine (SVM) was trained on geographical coordinates (Figure 4). Two clusters are visible: low mean life expectancy in the North and high mean life expectancy in the South, providing visual confirmation of life expectancy inequality.

Figure 4: Scatter graph – K-means threshold and SVM line



In Figure 5, income and employment deprivation follow the North-South divide. Education and health deprivation show high deprivation in the North, with medium to high deprivation in parts of the South and Southeast. Environmental quality and crime are more fragmented, reflecting urbanisation rather than a North-South pattern. This indicates that regional inequality arises from the accumulation of multiple deprivation sub-domains rather than a single factor.

Figure 5: Scatter graph – Spatial Distributions of the IMD Subdomains

A Welch's T-Test was performed on the K-means clustering of the data, confirming that life expectancy had a statistically significant difference ($p < 0.001$) between the North and South clusters, demonstrating that the disparity is a robust statistical reality.

Figure 6 reveals a strong negative linear relationship between deprivation scores and life expectancy, particularly income and employment. Education and crime have a weaker negative relationship in comparison, while the living environment score has nearly no correlation. Population_Density was log-transformed to reduce skewness and clearly visualise its correlation with Mean_LifeExp

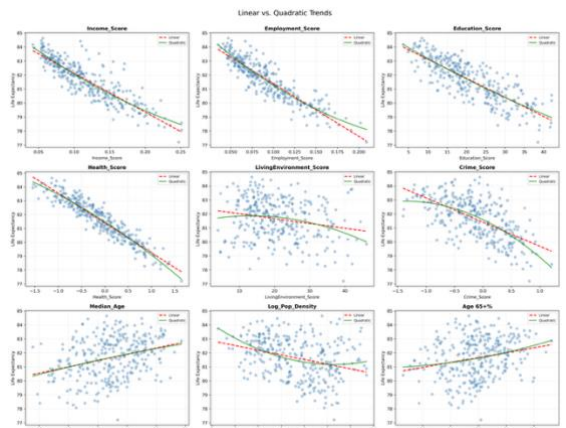


Figure 6: Scatter graph – Relationship between the independent and target variable

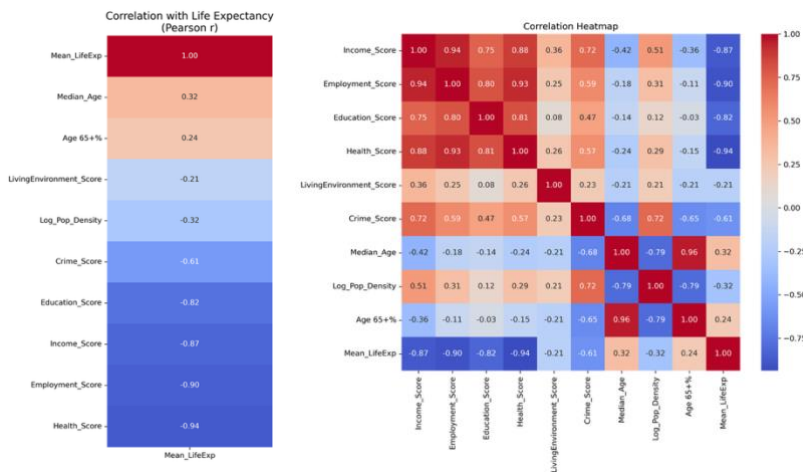


Figure 7 demonstrates extremely high levels of multicollinearity between Income and Employment Deprivation, indicating that they both function identically. The living environment score has the weakest correlation value of $r = -0.21$.

Figure 7: Correlation heatmap, and focused heatmap

Multicollinearity was tested and addressed by excluding Age 65+, LivingEnvironment_Score, as they were weakly correlated with Life Expectancy. Female and Male life expectancy variables were also dropped to create a parsimonious dataset. Income_Score and Employment_Score were highly correlated; therefore, to test which variable to keep, two separate multilinear regressions were executed.

3.3 Feature Engineering

The Median_Age was mean-centred, and all the IMD subdomains were Z-score normalised to ensure a stabilised dataset. Scaling the dataset meant the coefficients in the regression models could be directly compared. A Region_Group variable was created by grouping the 9 official regions into five distinct regions, allowing for analysis on a finer scale, capturing additional opportunities for detecting geographic trends while preserving enough sample size for maintaining statistical power in the modelling.

3.4 Explanatory Modelling

Figure 8 shows a Directed Acyclic Graph (DAG) that was developed to illustrate the expected relationships between the variables. The DAG shows that Region is the primary exposure influencing the socioeconomic characteristics that then serve as pathways leading to Health and Life Expectancy. The approach taken here considers both mediators and confounders in the model development rather than treating each variable as an independent predictor.

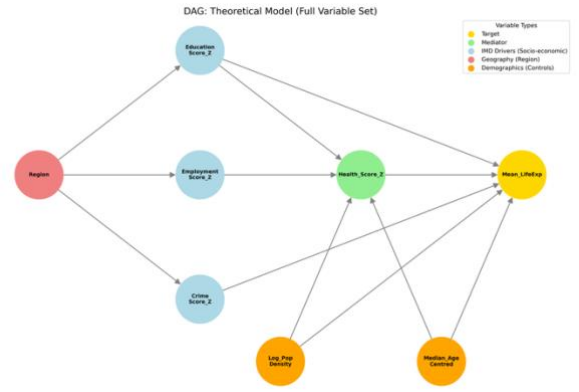


Figure 8: Directed Acyclic Graph (DAG)

Models 1 and 2:

$$\text{Mean_LifeExp} = 80.77 - 0.8789(\text{Income_Score_Z}) - 0.5007(\text{Education_Score_Z}) - 0.0693(\text{Crime_Score_Z}) + 0.1326(\text{Log_Pop_Density}) + 0.0225(\text{Median_Age_Centred}) + \epsilon$$

$$\text{Mean_LifeExp} = 80.63 - 1.0316(\text{Employment_Score_Z}) - 0.3191(\text{Education_Score_Z}) - 0.0164(\text{Crime_Score_Z}) + 0.1530(\text{Log_Pop_Density}) + 0.0756(\text{Median_Age_Centred}) + \epsilon$$

Two multiple linear regression models were compared to determine which model best explained Mean_LifeExp. The results showed that Model 2 containing Employment_Score was more robust and provided an improved adjusted R^2 of 0.868 compared to Model 1 ($R^2=0.826$).

Model 3:

$$\text{Mean_LifeExp} = 80.90 - 0.3697(\text{Employment_Score_Z}) - 0.813(\text{Health_Score_Z}) - 0.1785(\text{Education_Score_Z}) - 0.0643(\text{Crime_Score_Z}) + 0.1119(\text{Log_Pop_Density}) + 0.0489(\text{Median_Age_Centred}) + \epsilon$$

Implementing Health_Score_Z resulted in an improved adjusted R^2 of 0.906. However, the Employment_Score coefficient decreased dramatically from -1.03 to -0.37; this decrease demonstrates that it acts as a variable absorbing the effect of employment deprivation.

Model 4:

$$\text{Mean_LifeExp} = 81.29 - 0.3094(\text{Employment_Score_Z}) - 0.7236(\text{Health_Score_Z}) - 0.2157(\text{Education_Score_Z}) - 0.1060(\text{Crime_Score_Z}) + 0.0732(\text{Log_Pop_Density}) + 0.0427(\text{Median_Age_Centred}) - 0.3188(\text{North}) + \epsilon$$

The binary North variable was implemented, and it revealed that after accounting for the variations in demographics and deprivation, the North had a 0.32-year reduction in life expectancy in comparison to the South. The overall adjusted R^2 value slightly improved to 0.913. The model confirms that the North-South Divide is an additional and independent geographic influence on the outcome variable, in addition to deprivation.

Model 5:

$$\text{Mean_LifeExp} = 81.37 - 0.3698(\text{Employment_Score_Z}) - 0.6982(\text{Health_Score_Z}) - 0.1699(\text{Education_Score_Z}) - 0.0986(\text{Crime_Score_Z}) + 0.0476(\text{Log_Pop_Density}) + 0.0442(\text{Median_Age_Centred}) + 0.2903(\text{London}) - 0.2391(\text{Midlands}) - 0.22(\text{North}) + 0.0899(\text{South}) + \epsilon$$

The Region_Group variable was implemented, resulting in an adjusted R^2 value of 0.914, making it the strongest model so far. The disparity is still visible, with the Midlands and North having a mean life expectancy of 0.24 and 0.22 years less than the East. London has a mean life expectancy of 0.29 years more than the East.

3.5 Analysis, Interpretation, and Validation

Overall, while health deprivation is a significant predictor, employment is an underlying root cause, driving the disparity. Model 5 proves the existence of an ongoing Structural North-South divide, and that even when controlling for social factors, the geographic penalty of the North-South divide remains. In Figure 8, the residuals on the 45-degree line (Q-Q Plot) appear to align with our expectations that they are normally distributed (with only a small amount of deviation).

However, the Breusch-Pagan test results show heteroscedasticity ($p = 0.0415 < 0.05$), which was dealt with through the use of robust standard errors, to ensure that regional coefficients are still unbiased. Although the Employment VIF (10.8) is quite high due to many of the social variables being interrelated, this does not negatively impact our ability to identify independent geographic effects.

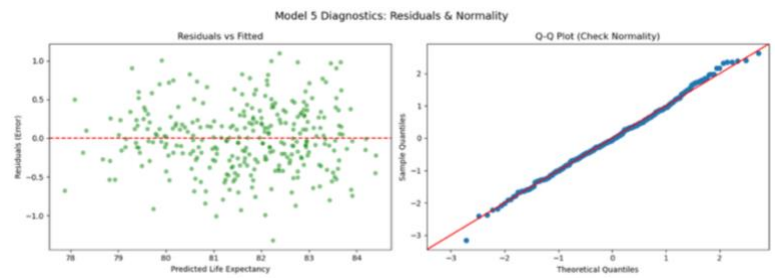


Figure 9: Residuals and Q-Q Plots for Model 5

4. Findings, Reflections and Future Work

The sequential approach to regression analysis resulted in a comprehensive view of how various factors influence life expectancy in England. Below are summaries of the results of each analytical question.

Q1:

The results of the t-test from the initial EDA analysis identified a significant difference ($p < 0.05$) in the average life expectancy between people living in the North and South. The geographic penalty described in Model 5 confirmed this finding, with both the North and Midlands losing an average of 0.22 years and 0.24 years, respectively, compared to the South at the same level of deprivation. This indicates that the life expectancy disparity is evidence of a structural issue rather than simply being an indicator of deprivation.

Q2:

The staged modelling clearly demonstrates a masking effect. `Employment_Score` appeared as the strongest predictor of life expectancy in Model 2 (coef = -1.0316), however, `Health_Score_Z` became the strongest immediate determinant of Model 3 (coef = -0.813). While health is more significant than either employment or education and crime, the path analyses demonstrate that it was through the effect of employment deprivation on health that life expectancy was affected. `Education_Score` and `Crime_Score` were the weakest socioeconomic predictors of life expectancy, while `Crime_Score` showed a negligible effect, indicating that unemployment is far more important than crime levels and educational attainment in driving life expectancy.

Q3:

Implementing a deeper geographical classification in Model 5 was a pivotal step in optimising model fit, which led to an adjusted R^2 value of 0.914. This demonstrates that using a more detailed geographic lens allows for the identification of structural inequality, which broader models tend to mask. Model 5 classified regions into distinct groupings, which demonstrated that there are significant independent predictors to explain 91.4% of the variance in life expectancy due to regional dissimilarities.

4.2 Reflections

The adjusted R^2 value of 0.914 indicates that IMD sub-domains are statistically robust predictors, and therefore a valid predictive tool. Employment is retained in the model despite a VIF of 10.8; despite its collinearity with the other variables, it must be kept in the model as it is necessary to maintain the model's causal integrity. The implementation of HC3 Robust Standard Errors was another limiting refinement in dealing with heteroscedasticity (Breusch-Pagan $p = 0.0415$) that confirms the statistical validity of the regional differences.

4.3 Limitations

Although the results of this report apply broadly across England, the heavy tails of the Q-Q Plot indicate that there are some outliers for the model performance at the extreme ends of the scale, particularly for high-deprivation districts. These outlier districts may be influenced by more local factors, such as industrial legacies, than by national socio-economic trends. While the structural drivers are adequately represented in this model, it does not include any lifestyle data (e.g., smoking, obesity), making it difficult to comment on lifestyle choices that occur in the structure. The model is thus better suited to determining structural policy and is not likely to be effective in determining localised clinical interventions.

4.4 Future Work

For future studies, the use of Spatial Error Models (SEM) is needed to capture regional spillover effects in health outcomes that go beyond administrative borders. The SEM addresses the possibility of spatial dependence in residuals that cannot be accounted for by using standard OLS models (Boulieri and Blangiardo, 2020). Longitudinal datasets will also address whether or not the disparity is still evident or changing.

5. Wordcount

Section 1: 257

Section 2: 277

Section 3: 978

Section 4: 566

6. References

Bambra, C., Munford, L. and Brown, H. (2018). *Health for Wealth: Building a Healthier Northern Powerhouse for UK Productivity*. [online] Available at:
<https://www.thenhsa.co.uk/app/uploads/2018/11/NHSA-REPORT-FINAL.pdf>.

Boulieri, A. and Blangiardo, M. (2020). Spatio-temporal model to estimate life expectancy and to detect unusual trends at the local authority level in England. *BMJ Open*, 10(11), p.e036855.
doi:<https://doi.org/10.1136/bmjopen-2020-036855>.

Google.com. (2019). *2019 Local Authority District Summaries lower-tier*. [online] Available at:
https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://assets.publishing.service.gov.uk/media/5d8b3cfbe5274a08be69aa91/File_10_-_IoD2019_Local_Authority_District_Summaries_lower-tier_.xlsx&ved=2ahUKEwixz8ysk9KQAxU0d0EAHQmiHCsQFnoECCEQAQ&usg=AOvVaw2e-DOB89oAbX5r0IZ1ayrk [Accessed 20 Dec. 2025].

Marmot, M., Allen, J., Boyce, T., Goldblatt, P. and Morrison, J. (2020). *HEALTH EQUITY IN ENGLAND: THE MARMOT REVIEW 10 YEARS ON*. [online] Available at:
https://www.health.org.uk/sites/default/files/upload/publications/2020/Health%20Equity%20in%20England_The%20Marmot%20Review%2010%20Years%20On_full%20report.pdf.

Office For National Statistics (2020). *Population estimates for the UK, England and Wales, Scotland and Northern Ireland, provisional - Office for National Statistics*. [online] www.ons.gov.uk. Available at:
<https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/bulletins/annualmidyearpopulationestimates/mid2019>.

Office for National Statistics (2021). *National life tables – life expectancy in the UK - Office for National Statistics*. [online] www.ons.gov.uk. Available at: <https://www.ons.gov.uk/peoplepopulationandcommunity/birthsdeathsandmarriages/lifeexpectancies/bulletins/nationallifetablesunitedkingdom/2018to2020>.

Office for National Statistics (2024). *Life expectancy for local areas of great britain*. [online] [Ons.gov.uk](http://ons.gov.uk). Available at: <https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/healthandlifeexpectancies/bulletins/lifeexpectancyforlocalareasoftheuk/between2001to2003and2021to2023>.

Ons.gov.uk. (2024). *Life expectancy for local areas in England, Northern Ireland and Wales: between 2001 to 2003 and 2020 to 2022*. [online] Available at: <https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/healthandlifeexpectancies/datasets/lifeexpectancyforlocalareasinenglandandnorthernirelandandwalesbetween2001to2003and2020to2022>.

Rutter, H., Savona, N., Glonti, K., Bibby, J., Cummins, S., Finegood, D.T., Greaves, F., Harper, L., Hawe, P., Moore, L., Petticrew, M., Rehfuss, E., Shiell, A., Thomas, J. and White, M. (2017). The Need for a Complex Systems Model of Evidence for Public Health. *The Lancet*, 390(10112), pp.2602–2604. doi:[https://doi.org/10.1016/s0140-6736\(17\)31267-9](https://doi.org/10.1016/s0140-6736(17)31267-9).

Statistics.gov.uk. (2019). *Local Authority Districts (December 2019) Boundaries UK BGC*. [online] Available at: <https://geoportal.statistics.gov.uk/datasets/local-authority-districts-december-2019-boundaries-uk-bgc/explore?location=51.770672%2C0.114979%2C8.46> [Accessed 20 Dec. 2025].

7. Appendix

In accordance with academic integrity guidelines, Google Gemini was utilised as a technical thought partner throughout the lifecycle of this project. The AI was not used to generate the project's original arguments or conclusions, but rather to assist in code optimisation, statistical validation, and methodological refinement.

The following prompts were used to assist in the technical execution of the project codes.

1. Geospatial Analysis & SVM Prompts

- "How do I extract the coordinates of the decision boundary from a Linear Support Vector Machine (SVM) to overlay a 'North-South line' on a latitude/longitude scatter plot?"
- "Explain the mathematical difference between using K-Means to find a 'Natural Divide' versus using a pre-defined 5-region administrative classification for life expectancy."
- "Provide the Python syntax for mapping a categorical variable like Geo_Location into a binary 0/1 variable for use in an OLS regression."

2. Statistical Diagnostics & Quality Scatter Plots

- "How can I generate a Grid of 6 Quality Scatter Plots where the color bar represents IMD deprivation scores and the axes represent Latitude/Longitude?"
- "Give me the code to plot a Q-Q Plot using `scipy.stats` to check for heavy-tailed residuals in my life expectancy model."

- "What is the best way to visualize a Residuals vs. Fitted plot to identify if I have heteroscedasticity in my data?"

3. Explanatory Modeling & Robustness

- "Explain the VIF (Variance Inflation Factor) threshold for Employment_Score_Z. If my VIF is 10.8, what are the theoretical justifications for keeping it in the model based on a Directed Acyclic Graph (DAG)?"
- "Provide the code for running a Breusch-Pagan test to statistically verify the presence of heteroscedasticity before applying robust errors."
- "How do I apply HC3 Robust Standard Errors specifically to an existing sm.OLS result object to ensure my 'North Penalty' coefficients are conservative?"

4. Causal Logic & Theoretical Models

- "How should I structure a DAG (Directed Acyclic Graph) to show that Education and Employment are upstream drivers that manifest through Health_Score_Z?"
- "Help me refine the logic for a Staged Modeling approach: why does the coefficient for Employment drop significantly when Health_Score_Z is introduced in Stage 3?"